

Signed-perspective coherence-preserving maps

Abstract

We introduce a signed-perspective class of coherence-preserving maps on finite gamble spaces. Given a strictly positive probability vector q and coordinatewise signed concave generators ρ_i , the score is

$$\Phi_i(g) = (q \cdot g) \rho_i \left(\frac{g_i}{q \cdot g} \right), \quad q \cdot g > 0.$$

The signed-concavity hypotheses imply continuity, positive homogeneity, coordinatewise sign preservation, and coherence preservation for pullbacks of every compact family of strictly desirable halfspaces. The proof is the same sublinearity argument as in the perspective-generator construction: for every output probability p , the scalar pullback $-p \cdot \Phi$ is a convex homogeneous sum of perspective terms. Injectivity is independent of the CPM property and is characterised by a projective injectivity condition on the slice $q \cdot t = 1$; a simple sufficient condition is strict monotonicity of every generator. The older perspective family is recovered as the special case $\rho_i(t) = t - \psi_i(t)$, where positive coordinates are copied exactly.

1 Setup

Fix $n \geq 2$ and identify the gamble space with $\mathbb{R}^n$. Let

$$\Delta^{n-1} := \left\{ p \in \mathbb{R}_{\geq 0}^n : \sum_i p_i = 1 \right\}, \quad \text{int } \Delta^{n-1} := \{ p \in \Delta^{n-1} : p_i > 0 \ \forall i \}.$$

For $p \in \Delta^{n-1}$ set

$$D_p := (\mathbb{R}_{\geq 0}^n \setminus \{0\}) \cup \{ h \in \mathbb{R}^n : p \cdot h > 0 \}.$$

For a nonempty $P \subseteq \Delta^{n-1}$ define $D_P := \bigcap_{p \in P} D_p$. A set $D \subseteq \mathbb{R}^n$ is a coherent set of strictly desirable gambles if:

- (D1) $\mathbb{R}_{\geq 0}^n \setminus \{0\} \subseteq D$;
- (D2) $D \cap \mathbb{R}_{\leq 0}^n = \emptyset$;
- (D3) if $g, h \in D$ and $\lambda, \mu > 0$, then $\lambda g + \mu h \in D$;
- (D4) if $g \in D \setminus \mathbb{R}_{\geq 0}^n$, then there exists $\delta > 0$ such that $g - \delta \mathbf{1} \in D$.

2 Signed concave generators

Definition 2.1 (Signed concave generator family). Fix $q \in \text{int } \Delta^{n-1}$. A family $\rho = (\rho_1, \dots, \rho_n)$ is a signed concave generator family if, for each i ,

- (S1) $\rho_i : \mathbb{R} \rightarrow \mathbb{R}$ is finite and concave;
- (S2) $\rho_i(0) = 0$;
- (S3) $\rho_i(t) > 0$ for $t > 0$, and $\rho_i(t) < 0$ for $t < 0$.

Since each ρ_i is finite and concave on $\mathbb{R}$, it is continuous. Define the open halfspace

$$G_q^+ := \{g \in \mathbb{R}^n : q \cdot g > 0\}.$$

The signed-perspective score associated with (q, ρ) is the map

$$\Phi_i(g) := (q \cdot g) \rho_i\left(\frac{g_i}{q \cdot g}\right), \quad g \in G_q^+, \quad i = 1, \dots, n.$$

All preimages below are domain-relative: for $A \subseteq \mathbb{R}^n$,

$$\Phi^{-1}(A) := \{g \in G_q^+ : \Phi(g) \in A\}.$$

3 Basic properties

Lemma 3.1 (Continuity, homogeneity, and sign preservation). *The map $\Phi : G_q^+ \rightarrow \mathbb{R}^n$ is continuous and positively homogeneous of degree one. Moreover, for every $g \in G_q^+$ and every coordinate i ,*

$$\text{sgn } \Phi_i(g) = \text{sgn } g_i.$$

In particular,

$$\Phi(G_q^+) \subseteq \mathbb{R}^n \setminus \mathbb{R}_{\leq 0}^n.$$

Also $\mathbb{R}_{\geq 0}^n \setminus \{0\} \subseteq G_q^+$ and Φ sends every nonzero partial gain to a nonzero partial gain with the same zero pattern.

Proof. Continuity follows from continuity of the ρ_i and positivity of $q \cdot g$ on G_q^+ . If $c > 0$, then

$$q \cdot (cg) = c(q \cdot g), \quad \frac{(cg)_i}{q \cdot (cg)} = \frac{g_i}{q \cdot g},$$

so $\Phi_i(cg) = c\Phi_i(g)$. Since $q \cdot g > 0$, the sign of $\Phi_i(g)$ is the sign of $\rho_i(g_i/(q \cdot g))$, which by (S2)–(S3) is exactly the sign of g_i . If $g \in G_q^+$, then some coordinate of g is positive, hence the corresponding coordinate of $\Phi(g)$ is positive; so $\Phi(g) \notin \mathbb{R}_{\leq 0}^n$. Finally, if $g \in \mathbb{R}_{\geq 0}^n \setminus \{0\}$, then $q \cdot g > 0$ because $q \in \text{int } \Delta^{n-1}$, and the sign assertion gives $\Phi(g) \in \mathbb{R}_{\geq 0}^n \setminus \{0\}$ with precisely the same zero coordinates as g . $\square$

4 The CPM theorem

For $p \in \Delta^{n-1}$ define

$$G_p(g) := -p \cdot \Phi(g) = -\sum_{i=1}^n p_i (q \cdot g) \rho_i\left(\frac{g_i}{q \cdot g}\right), \quad g \in G_q^+.$$

Lemma 4.1 (Sublinearity of the scalar pullbacks). *For every $p \in \Delta^{n-1}$, the function $G_p : G_q^+ \rightarrow \mathbb{R}$ is convex and positively homogeneous of degree one. Consequently it is subadditive on G_q^+ :*

$$G_p(g+h) \leq G_p(g) + G_p(h), \quad g, h \in G_q^+.$$

Proof. Set $\eta_i := -\rho_i$. Since ρ_i is concave, η_i is convex. The perspective

$$(u, s) \longmapsto s\eta_i(u/s), \quad s > 0,$$

is convex on $\mathbb{R} \times \mathbb{R}_{>0}$. Composing it with the affine map

$$g \longmapsto (g_i, q \cdot g)$$

gives convexity of

$$g \mapsto (q \cdot g) \eta_i \left(\frac{g_i}{q \cdot g} \right)$$

on G_q^+ . Since $p_i \geq 0$, G_p is a nonnegative weighted sum of these convex terms. Thus G_p is convex. Positive homogeneity follows directly from the definition of Φ and the positive homogeneity of Φ . A convex positively homogeneous function of degree one is subadditive: for $g, h \in G_q^+$,

$$G_p(g + h) = 2G_p\left(\frac{g + h}{2}\right) \leq 2\left(\frac{G_p(g)}{2} + \frac{G_p(h)}{2}\right) = G_p(g) + G_p(h).$$

□

For $p \in \Delta^{n-1}$ put

$$K_p := \{g \in G_q^+ : G_p(g) < 0\} = \{g \in G_q^+ : p \cdot \Phi(g) > 0\}.$$

Lemma 4.2 (Single-halfspace pullback). *For every $p \in \Delta^{n-1}$,*

$$\Phi^{-1}(D_p) = (\mathbb{R}_{\geq 0}^n \setminus \{0\}) \cup K_p.$$

Proof. If $g \in \mathbb{R}_{\geq 0}^n \setminus \{0\}$, then $g \in G_q^+$ and $\Phi(g) \in \mathbb{R}_{\geq 0}^n \setminus \{0\}$, so $g \in \Phi^{-1}(D_p)$. If $g \notin \mathbb{R}_{\geq 0}^n$ and $g \in G_q^+$, then some coordinate of g is negative; by strict sign preservation the same coordinate of $\Phi(g)$ is negative, so $\Phi(g)$ is not a partial gain. Hence $\Phi(g) \in D_p$ iff $p \cdot \Phi(g) > 0$, which is exactly $g \in K_p$. Points outside G_q^+ are outside the domain of Φ and hence outside the preimage. □

Theorem 4.3 (Signed-perspective CPM theorem). *Let $q \in \text{int } \Delta^{n-1}$ and let ρ be a signed concave generator family. For every nonempty compact set $P \subseteq \Delta^{n-1}$, the pullback*

$$\Phi^{-1}(D_P) = \bigcap_{p \in P} \Phi^{-1}(D_p)$$

is a coherent set of strictly desirable gambles.

Proof. By the previous lemma,

$$\Phi^{-1}(D_P) = (\mathbb{R}_{\geq 0}^n \setminus \{0\}) \cup \bigcap_{p \in P} K_p.$$

We verify the four axioms.

(D1) If $g \in \mathbb{R}_{\geq 0}^n \setminus \{0\}$, then g belongs to the displayed partial-gain part.

(D2) If $g \in \mathbb{R}_{\leq 0}^n$, then $q \cdot g \leq 0$ because $q > 0$. Thus $g \notin G_q^+$. Also $g \notin \mathbb{R}_{\geq 0}^n \setminus \{0\}$, either because $g = 0$ or because g has a negative coordinate. Hence $g \notin \Phi^{-1}(D_P)$.

(D3) Let $g, h \in \Phi^{-1}(D_P)$ and let $\lambda, \mu > 0$. If both g, h are nonzero partial gains, then $\lambda g + \mu h$ is a nonzero partial gain. If neither is a partial gain, then $g, h \in K_p$ for every $p \in P$. By positive homogeneity and subadditivity of each G_p ,

$$G_p(\lambda g + \mu h) \leq \lambda G_p(g) + \mu G_p(h) < 0,$$

so $\lambda g + \mu h \in K_p$ for every $p \in P$. Finally suppose, without loss of generality, that $g \in \mathbb{R}_{\geq 0}^n \setminus \{0\}$ and $h \in K_p$ for every $p \in P$. Then $\Phi(g) \in \mathbb{R}_{\geq 0}^n \setminus \{0\}$, so $G_p(g) = -p \cdot \Phi(g) \leq 0$ for every $p \in P$. Hence

$$G_p(\lambda g + \mu h) \leq \lambda G_p(g) + \mu G_p(h) < 0,$$

for every $p \in P$. Thus $\lambda g + \mu h \in \bigcap_{p \in P} K_p$.

(D4) Let $g \in \Phi^{-1}(D_P) \setminus \mathbb{R}_{\geq 0}^n$. Then $g \in K_p$ for every $p \in P$. In particular $q \cdot g > 0$ and $G_p(g) < 0$ for every $p \in P$. Since $p \mapsto G_p(g)$ is continuous and P is compact,

$$M := \max_{p \in P} G_p(g) < 0.$$

Choose $\delta_0 \in (0, q \cdot g)$. For $0 \leq \delta \leq \delta_0$,

$$q \cdot (g - \delta \mathbf{1}) = q \cdot g - \delta \geq q \cdot g - \delta_0 > 0,$$

so $g - \delta \mathbf{1} \in G_q^+$. The map

$$(p, \delta) \mapsto G_p(g - \delta \mathbf{1})$$

is continuous on the compact set $P \times [0, \delta_0]$. Hence it is uniformly continuous. Choose $\delta \in (0, \delta_0]$ small enough that

$$|G_p(g - \delta \mathbf{1}) - G_p(g)| < -M/2 \quad \forall p \in P.$$

Then

$$G_p(g - \delta \mathbf{1}) \leq G_p(g) - M/2 \leq M/2 < 0 \quad \forall p \in P.$$

Thus $g - \delta \mathbf{1} \in \bigcap_{p \in P} K_p \subseteq \Phi^{-1}(D_P)$. □

5 Injectivity

The CPM theorem above does not require injectivity. Injectivity is a separate projective condition. The next proposition gives the exact criterion.

Let

$$\Sigma_q := \{t \in \mathbb{R}^n : q \cdot t = 1\}, \quad R_\rho(t) := (\rho_1(t_1), \dots, \rho_n(t_n)).$$

Since every $g \in G_q^+$ has a unique decomposition

$$g = st, \quad s := q \cdot g > 0, \quad t := g/(q \cdot g) \in \Sigma_q,$$

we have

$$\Phi(g) = sR_\rho(t).$$

Definition 5.1 (Projective injectivity on the q -slice). The generator family ρ is projectively injective on Σ_q if, whenever $t, u \in \Sigma_q$ and $a > 0$ satisfy

$$R_\rho(t) = aR_\rho(u),$$

then $a = 1$ and $t = u$.

Proposition 5.2 (Injectivity criterion). *The signed-perspective score $\Phi : G_q^+ \rightarrow \mathbb{R}^n$ is injective if and only if ρ is projectively injective on Σ_q .*

Proof. First assume that ρ is projectively injective. Suppose $\Phi(g) = \Phi(h)$. Write

$$g = st, \quad h = s'u, \quad s = q \cdot g > 0, \quad s' = q \cdot h > 0, \quad t, u \in \Sigma_q.$$

Then

$$sR_\rho(t) = s'R_\rho(u),$$

so

$$R_\rho(t) = (s'/s)R_\rho(u).$$

Projective injectivity gives $s'/s = 1$ and $t = u$. Hence $g = h$.

Conversely, suppose projective injectivity fails. Then there exist $t, u \in \Sigma_q$ and $a > 0$ such that

$$R_\rho(t) = aR_\rho(u)$$

but not both $a = 1$ and $t = u$. Put $g := t$ and $h := au$. Since $q \cdot t = q \cdot u = 1$, both $g, h \in G_q^+$ and

$$\Phi(g) = R_\rho(t) = aR_\rho(u) = \Phi(au) = \Phi(h).$$

Moreover $g \neq h$: if $t = au$, then applying $q \cdot$ gives $1 = a$, and then $t = u$, contrary to the assumed failure. Hence Φ is not injective. $\square$

Lemma 5.3 (A concrete sufficient condition for injectivity). *If each ρ_i is strictly increasing on $\mathbb{R}$, then ρ is projectively injective on Σ_q . Hence Φ is injective.*

Proof. Suppose $R_\rho(t) = aR_\rho(u)$ for some $t, u \in \Sigma_q$ and $a > 0$. If $a = 1$, strict increase of each ρ_i gives $t_i = u_i$ for every i , hence $t = u$.

Assume $a > 1$. We claim that $t_i \geq au_i$ for every i . If $u_i = 0$, then $\rho_i(t_i) = a\rho_i(0) = 0$, and sign preservation gives $t_i = 0 = au_i$. If $u_i \neq 0$, concavity and $\rho_i(0) = 0$ imply

$$\rho_i(au_i) \leq a\rho_i(u_i).$$

Since $\rho_i(t_i) = a\rho_i(u_i)$ and ρ_i is strictly increasing, it follows that $t_i \geq au_i$. Therefore

$$1 = q \cdot t \geq a(q \cdot u) = a > 1,$$

a contradiction. If $0 < a < 1$, interchange t and u and apply the previous argument to $1/a > 1$. Thus $a = 1$, and then $t = u$ as above. $\square$

Corollary 5.4 (Homeomorphism onto the image). *If ρ is projectively injective on Σ_q , then $\Phi : G_q^+ \rightarrow \Phi(G_q^+)$ is a homeomorphism.*

Proof. The domain G_q^+ is open in $\mathbb{R}^n$, and Φ is continuous and injective. By invariance of domain, $\Phi(G_q^+)$ is open in $\mathbb{R}^n$ and the inverse map from $\Phi(G_q^+)$ to G_q^+ is continuous. $\square$

Remark 5.5 (What is, and is not, automatic). Concavity and sign preservation of the one-dimensional generators are enough for the CPM theorem, but they are not the same thing as injectivity. For publication-level use, injectivity should therefore either be verified directly for the chosen generator family or imposed through the projective injectivity condition above. In the inverse-quadratic case, and in the older perspective setting when an explicit scalar inversion theorem is available, injectivity is proved by recovering the scale $q \cdot g$ from a one-dimensional equation such as $F(S) = B$; that special scalar recovery mechanism is absent from the present signed-perspective framework. The CPM proof and the injectivity proof are logically independent: the former is a convexity/sublinearity argument for all scalar pullbacks $-p \cdot \Phi$, while the latter is a projective one-to-one condition on the slice Σ_q .

6 Examples

Example 6.1 (Injective signed-diagonal scores). Fix constants $c_i^-, c_i^+ > 0$ with $c_i^- \geq c_i^+$, and define

$$\rho_i(t) := \begin{cases} c_i^+ t, & t \geq 0, \\ c_i^- t, & t \leq 0. \end{cases}$$

Then ρ_i is finite, concave, strictly increasing, and signed. Hence the associated signed-perspective score is a CPM and is injective by the sufficient condition above. Explicitly,

$$\Phi_i(g) = \begin{cases} c_i^+ g_i, & g_i \geq 0, \\ c_i^- g_i, & g_i \leq 0. \end{cases}$$

This example is already outside the older perspective family whenever some $c_i^+ \neq 1$, because positive coordinates are not copied exactly. Taking $c_i^- > c_i^+ > 0$ gives a genuinely signed-diagonal concave kink at the seam.

Example 6.2 (Noninjective signed concave generator). Let $n \geq 3$, let $q = (1/n, \dots, 1/n)$, and set

$$\rho_i(t) := \min\{t, 1\} \quad (i = 1, \dots, n).$$

This function is finite, concave, satisfies $\rho_i(0) = 0$, and is strictly sign-preserving, but it is not strictly increasing on $[1, \infty)$. For $n = 3$, put

$$t = \left(\frac{6}{5}, \frac{6}{5}, \frac{3}{5}\right), \quad u = \left(\frac{13}{10}, \frac{11}{10}, \frac{3}{5}\right).$$

Both points lie in Σ_q , they are distinct, and

$$R_\rho(t) = R_\rho(u) = \left(1, 1, \frac{3}{5}\right).$$

Thus projective injectivity fails, and so the associated Φ is not injective. This example shows that the injectivity condition is not vacuous.

7 Relation to the older perspective family

The earlier score

$$\phi_i(g) = g_i - (q \cdot g) \psi_i\left(\frac{g_i}{q \cdot g}\right)$$

is the special case

$$\rho_i(t) = t - \psi_i(t).$$

If each ψ_i is convex, nonnegative, and vanishes on $[0, \infty)$, then ρ_i is concave, $\rho_i(t) = t$ for $t \geq 0$, and $\rho_i(t) \leq t < 0$ for $t < 0$. Thus the older family is a signed-perspective family with the additional restriction that positive coordinates are copied exactly. The present theorem removes that restriction: positive coordinates may be transformed nontrivially, as long as the signed concavity hypotheses are preserved.

8 Status of exact representation of arbitrary compact credal sets

The CPM theorem should be separated from the stronger exact-representation question. Given a compact set $P \subseteq \text{int } \Delta^{n-1}$, exact representation by a single output halfspace would ask for some $q \in \text{int } \Delta^{n-1}$, some $p \in \Delta^{n-1}$, and some signed concave generators ρ_i such that

$$\Phi^{-1}(D_p) = D_P.$$

On the projective slice Σ_q , this is equivalent to the additive-separable sign representation

$$\sum_i p_i \rho_i(t_i) > 0 \iff \min_{r \in P} r \cdot t > 0, \quad t \in \Sigma_q.$$

The theorem above does not prove that such a representation exists for every compact interior P . There are structural reasons to treat the universal claim with caution: the left-hand side is additively separable in the coordinates on the slice, whereas the right-hand side is the lower envelope of an arbitrary compact family of affine functionals. This separability imposes restrictions, especially on the geometry of smooth boundary pieces and on the combinatorics of polyhedral examples. No universal representation theorem is claimed here.